



TRAINING PROGRAMME

Network Security

Secure network devices, firewalls, VPNs and access from the edge to the endpoint

51 HOURS

FOR Network administrators and engineers responsible for securing networks



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Overview

This programme teaches network engineers to secure the network itself. Participants harden device access and management, centralise authentication with AAA, filter traffic with ACLs, then deploy stateful, zone-based and next-generation firewalls, with a vendor-neutral lab on pfSense or OPNsense, and intrusion prevention. They secure endpoints and Layer 2, apply cryptography, build site-to-site and remote-access VPNs including WireGuard, compare them with zero trust network access, and test their defences. Every chapter ends with a lab, and the programme closes with a final project in which each team secures and validates a branch network.

At a glance

#	PHASE	FOCUS	HOURS
01	Securing Network Devices	Threats and mitigation, secure management, AAA, ACLs	12 h
02	Firewalls & IPS	Stateful and zone-based firewalls, NGFW, pfSense or OPNsense, Suricata, IPS	12 h
03	Endpoint, Layer 2 & Cryptography	Endpoint protection, Layer 2 attacks and mitigation, integrity, authenticity, PKI	6 h
04	VPNs, Zero Trust & Testing	Site-to-site IPsec, remote access with SSL and WireGuard, ZTNA, security testing	9 h
05	Final Project	A secured branch network, validated by tests	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		51 h

Key facts

AUDIENCE	Network administrators and engineers responsible for securing networks
PREREQUISITES	Switching and routing (course: Switching, Routing & Wireless Essentials (CCNA 2)), or equivalent. These are not taught again
DURATION	51 h: 45 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs on simulated Cisco devices and open-source firewalls
LAB ENVIRONMENT	Runs locally: one laptop per participant (16 GB RAM) with Cisco Packet Tracer (free with each participant's Cisco Networking Academy account) and pfSense or OPNsense and Linux VMs prepared by the trainer; isolated lab network; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase and a final project with validation tests; certificate of completion for participants who complete both
OUTCOME	Participants can harden network devices, deploy AAA, ACLs, firewalls, IPS, Layer 2 protections and VPNs, and prove the result with security tests

PHASE 01

12 H

Securing Network Devices

Participants identify what needs protecting in a network, secure administrative access and management protocols, centralise authentication with RADIUS and TACACS+, and design and apply access control lists. Validated lab: routers managed only over SSH with AAA through a server, sending NTP-synchronised logs to a syslog server, and ACLs that enforce the security policy.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Securing Networks	Assets, threats, mitigation	Lecture + Workshop	3 h
1.2	Device Access & Management	Secure access, administrative roles, SNMP, syslog, NTP	Lecture + Lab	3 h
1.3	AAA	RADIUS, TACACS+	Lecture + Lab	3 h
1.4	ACLs	Design and implementation	Lecture + Lab	3 h

PHASE 02

12 H

Firewalls & IPS

Participants build stateful and zone-based policy firewalls, learn what next-generation firewalls add, with application control, URL filtering, TLS inspection and Cisco Secure Firewall concepts, study IPS technologies and signatures, then deploy a vendor-neutral firewall on pfSense or OPNsense with NAT, VLANs and Suricata.

Validated lab: a pfSense or OPNsense firewall that segments three VLANs and blocks a test attack with Suricata.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Firewall Technologies	Stateful firewalls, zone-based policy firewalls	Lecture + Lab	3 h
2.2	Next-Generation Firewalls	Application control, IPS, URL filtering, TLS inspection; Cisco Secure Firewall concepts	Lecture + Workshop	3 h
2.3	IPS	IPS technologies, signatures, implementation	Lecture + Lab	3 h
2.4	pfSense or OPNsense Lab	Rules, NAT, VLANs, IPS with Suricata	Lab	3 h

PHASE 03

6 H

Endpoint, Layer 2 & Cryptography

Participants protect endpoints and the switched network against Layer 2 attacks, then apply the cryptography that VPNs and management protocols rely on: hashing, signatures and certificates. Validated lab: a switch configured with port security, DHCP snooping and dynamic ARP inspection, verified with show commands, and a certificate issued by the lab certificate authority.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Endpoint & Layer 2 Security	Endpoint protection, Layer 2 attacks and mitigation	Lecture + Lab	3 h
3.2	Cryptography	Integrity, authenticity, PKI	Lecture + Lab	3 h

PHASE 04

9 H

VPNs, Zero Trust & Testing

Participants connect sites with IPsec VPNs, give remote users access with IPsec, SSL VPN and WireGuard, compare these with zero trust network access, then test the network's defences with Nmap. Validated lab: two sites joined by an IPsec VPN, a remote user connected over WireGuard, and a scan report that confirms only the intended services are exposed.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Site-to-Site VPNs	IPsec concepts and configuration	Lecture + Lab	3 h
4.2	Remote-Access VPNs	IPsec and SSL VPNs, WireGuard	Lecture + Lab	1.5 h
4.3	Zero Trust Network Access	ZTNA principles and products	Lecture + Workshop	1.5 h
4.4	Security Testing	Testing techniques and tools	Lecture + Lab	3 h

PHASE 05

6 H

Final Project

Teams of two or three secure the network of a branch office from a written brief, reusing the phase labs, then prove their work with validation tests in front of the group.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Build	Read the brief, then deploy AAA, ACLs, a zone-based firewall or pfSense, IPS and Layer 2 security	Project	3 h
5.2	VPN & Validation	Site-to-site IPsec VPN to head office; validation tests (show commands and simulation in Packet Tracer, Nmap against the pfSense or OPNsense segment)	Project	1.5 h
5.3	Demo & Review	Live tests in front of the group; questions and feedback	Project	1.5 h

FINAL PROJECT

Securing a Branch Network

A branch network secured end to end: AAA for device access, ACLs, a zone-based firewall or pfSense with IPS, Layer 2 protections and a site-to-site IPsec VPN to head office, each control proved by a validation test.

STACK Cisco Packet Tracer, Cisco IOS, RADIUS or TACACS+, pfSense or OPNsense, Suricata, IPsec, Nmap

DELIVERABLE Configurations, network diagram, validation test plan and results, live demo